

CIE: Internal Assessment Details

Internal Assessment Question Paper – 1
Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of Computer Science and Engineering

Programme: B.E**Term: Oct 2021 - Feb 2022****Max Marks: 30****CIE: Test 1****Subject: Artificial Intelligence****Subject Code: CS53****Credits: 3:0:1****Sem: V****Sec: A, B, C****Date: 07/12/2021****Time: 10.30AM – 11:30AM****Portions for Test: (L1-L12)****Instructions to Candidates:**

1. **Question 1 is Compulsory.** Answer any one question from 2 and 3.
2. Each Question carries 15M. **Write figures wherever necessary.**
3. Mobiles, smart watches or any electronic gadgets are strictly banned.

Sl #	Question	Marks	Bloom's Level	CO Mapping
1	a. Discuss the foundations of AI.	04	Understand	CO1
	b. Illustrate the PEAS description for i) Taxi driver (ii) Medical diagnosis system	04	Apply	CO1
	c. Describe uninformed search techniques in detail with examples. Compare the performance metric values.	07	Understand	CO1
2	a. Discuss A* Algorithm. Find out an optimal to travel from one corner to another corner of a country using heuristic function (Straight-line distance between each node to the Goal node M). Here start node is E and Goal node is M. Fig. Road map of a country (Distances CB=38, LN=49)	08	Apply	CO1

heuristic

Act

178
38
118
238

P.T.O.

	Straight-line distance between a node to a goal node M <table><tr><td>A 24</td><td>F 145</td><td>K 108</td><td>P 65</td></tr><tr><td>B 28</td><td>G 123</td><td>L 51</td><td>Q 11</td></tr><tr><td>C 70</td><td>H 95</td><td>N 58</td><td></td></tr><tr><td>D 95</td><td>I 77</td><td>O 10</td><td></td></tr><tr><td>E 118</td><td>J 57</td><td>M 0</td><td></td></tr></table>	A 24	F 145	K 108	P 65	B 28	G 123	L 51	Q 11	C 70	H 95	N 58		D 95	I 77	O 10		E 118	J 57	M 0				
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	b. With a neat diagram explain agents interact with environment through sensors and actuators.	04	Understand	CO1																				
	c. Define Artificial Intelligence. List out any two applications of AI.	03	Understand	CO1																				
3	a. Explain the structure of a model based reflex agent with the help of a diagram.	04	Understand	CO1																				
	b. Discuss the properties of Best First Search.	04	Understand	CO1																				
	c. Describe Greedy BFS informed search techniques in detail with examples. Compare the performance metric values.	07	Understand	CO1																				

Course Outcomes meant to be assessed by the IA Test 1:

CO1: Identify the modern view of artificial intelligence and its applications based on agent philosophy and understand the various AI search strategies, their performance and applications using any programming environment. (PO-1,2,3,5,9,10, PSO-2,3)

CO2: Demonstrate an understanding of knowledge based agents and various knowledge representation and inference techniques and analyze how a particular inference technique works on a problem specification. (PO-2,3,4, PSO-2)